THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Sound is produced by: 1

(a) heat **(b)** light **(c)** vibration **(d)** rest

2. Ultrasonic sounds have frequency: 1

(a) below 20 Hz **(b)** 20 Hz to 20 kHz **(c)** exactly 20 Hz **(d)** above 20 kHz

3. Frequency 100 Hz means: 1

(a) 100 m **(b)** 100 oscillations per second

(c) 100 s per oscillation **(d)** 100 J

4. Sound needs a: 1

(a) magnet **(b)** vacuum **(c)** light **(d)** medium

5. Infrasonic sounds have frequency: 1

(a) below 20 Hz **(b)** 100 Hz **(c)** above 20 kHz **(d)** 20 kHz

6. Time period of 10 Hz is: 1

(a) 1 s **(b)** 100 s **(c)** 0.1 s **(d)** 10 s

7. Sound cannot travel in: 1

(a) steel **(b)** air **(c)** vacuum **(d)** water

8. Echo is due to: 1

(a) refraction **(b)** dispersion

(c) reflection of sound **(d)** absorption

9. Speed = frequency x: 1

(a) mass **(b)** amplitude **(c)** time **(d)** wavelength

10. Sound in air is a: 1

(a) longitudinal wave **(b)** transverse wave

(c) light wave **(d)** radio wave

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is sound? 2

12. What is wavelength? 2

13. Name two uses of ultrasound. 2

14. What is a wave? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three examples of sources of sound. 3

16. Find the time period of 250 Hz. 3

17. State the effect of temperature on the speed of sound. 3

OR

State three differences between a longitudinal and a transverse wave. 3

18. Name three uses of echo. 3

OR

What are infrasonic and ultrasonic sounds? Give one example each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how sound is produced and propagated. 5

OR

Explain the audible range and the infrasonic and ultrasonic sounds. 5

20. Describe the terms of a wave: crest, trough, amplitude, wavelength, frequency. 5

OR

Explain how a bat uses echolocation. 5

END OF QUESTION PAPER Total: 40 Marks

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